Single qubit operations: $V = \mathbb{C}^2$

$|\psi\rangle = a|0\rangle + b|1\rangle \quad a, b \in \mathbb{C}$

$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$|0\rangle, |1\rangle$ are eigenbasis of Z operator.

$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ Hadamard

$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

$H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

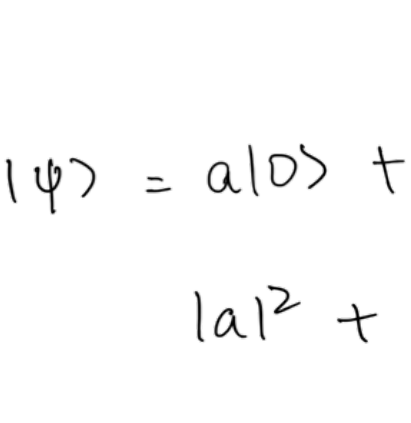
S-gate $S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix}$

T-gate $T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$

$S = T^2$

$|\psi\rangle = a|0\rangle + b|1\rangle \quad a, b \in \mathbb{C}$

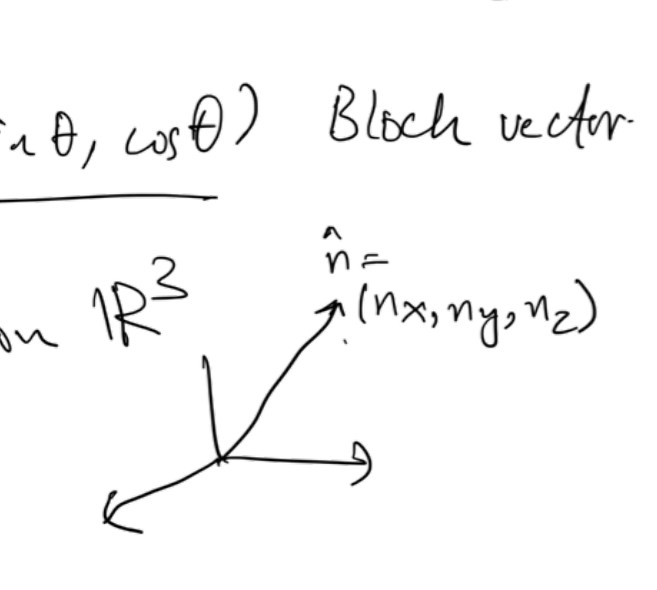
$|a|^2 + |b|^2 = 1 \Rightarrow$ we need three real numbers



$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\varphi} \sin\left(\frac{\theta}{2}\right)|1\rangle$

$\underline{(\cos\varphi \sin\theta, \sin\varphi \sin\theta, \cos\theta)}$ Bloch vector

In 3D J_x, J_y, J_z real rotation $\mathbb{R}^3$



$R = \exp(i\theta\, \hat{n} . \vec{J})$

$\hat{n} = (n_x, n_y, n_z)$

Rotation by an angle θ about the axis defined by $\hat{n}$.

$\sigma_x, \sigma_y, \sigma_z \Leftrightarrow X, Y, Z$

$R_x(\theta) = \exp\left(-\frac{i\theta X}{2}\right) = \cos\frac{\theta}{2} I - i\sin\left(\frac{\theta}{2}\right) X$

$R_y(\theta) = \exp\left(-i\frac{\theta}{2} Y\right) = \cos\left(\frac{\theta}{2}\right) I - i\sin\left(\frac{\theta}{2}\right) Y$

$R_z(\theta) = \exp\left(-i\frac{\theta}{2} Z\right) = \cos\frac{\theta}{2} I - i\sin\frac{\theta}{2} Z$

$R_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{+i\theta/2} \end{pmatrix}$

Similarly using $\mathbb{R}^3$ rotations, we can define an arbitrary rotation

$R_{\hat{n}}(\theta) = \exp\left(i\frac{\theta}{2}\, \hat{n} . \vec{\sigma}\right)$

$\hat{n} . \vec{\sigma} = n_x X + n_y Y + n_z Z$

$= \cos\left(\frac{\theta}{2}\right) I - i\sin\left(\frac{\theta}{2}\right)(n_x X + n_x Y + n_z Z)$

A generic single qubit unitary $U \in \underline{SU(2)}$

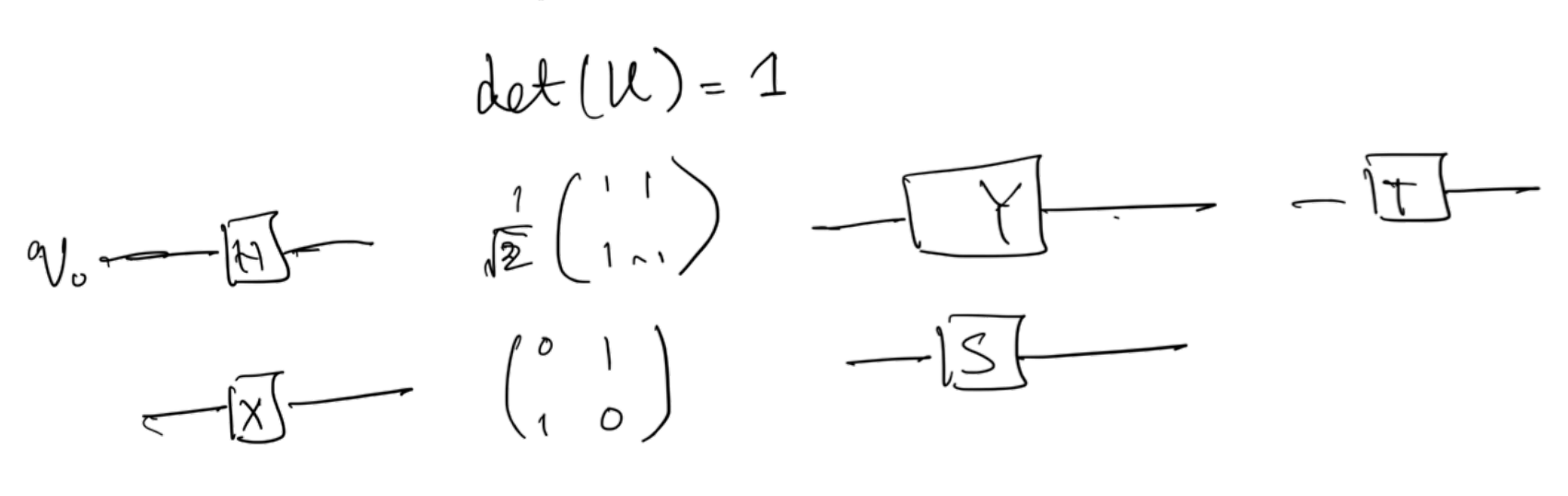
$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta)$

β, γ, δ are Euler angles.

SU(2): group of unitary transformations (2×2 complex matrices) given by $U = \exp(i\theta\, \hat{n} . \vec{\sigma})$

$U^\dagger U = I$.

$\det(U) = 1$



$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$

$R_x(\theta) = \exp\left(-i\frac{\theta}{2} X\right)$

Two qubit operations:

Controlled NOT gate:

if A is true then do B

CNOT ≡ CX : takes two qubits as inputs. control qubit, target qubit.

$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$

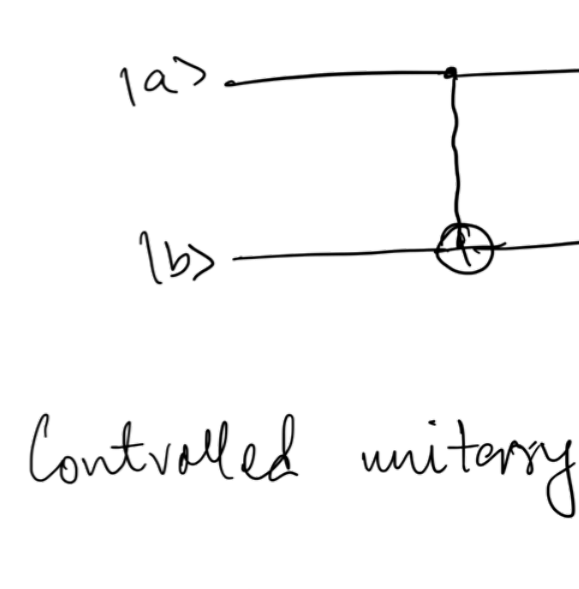
$= P_0 \otimes I + \underline{P_1} \otimes X.$

$X|0\rangle = |1\rangle$

$X|1\rangle = |0\rangle$

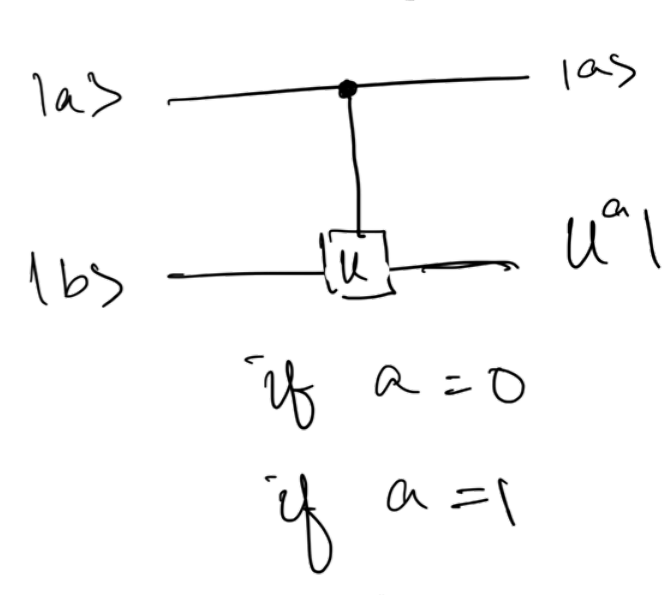
$P_0 = |0\rangle\langle 0|$

$P_1 = |1\rangle\langle 1|$



	a	b	a+b
	0	0	0
	0	1	1
→	1	1	0
→	1	0	1

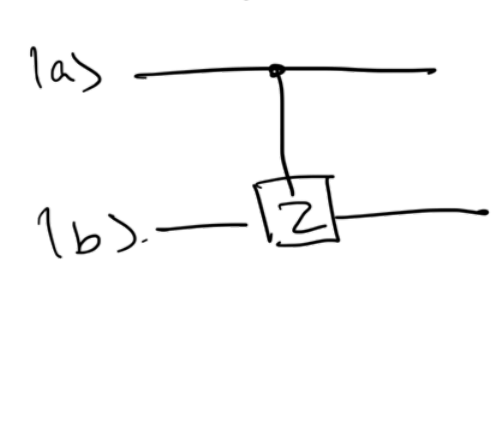
Controlled unitary.



if $a = 0$ $\quad U^0 = I$

if $a = 1$ $\quad U^1 = U$

Controlled-Z gate



$CZ = P_0 \otimes I + P_1 \otimes Z$

in the basis

$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$

$CZ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$

$\underline{H^\dagger Z H = X}$: Identity

Pauli-X $\left.\begin{aligned} |+\rangle &= \frac{|0\rangle + |1\rangle}{\sqrt{2}} \\ |-\rangle &= \frac{|0\rangle - |1\rangle}{\sqrt{2}} \end{aligned}\right\}$ Eigenvectors of X

$H(CZ)H = \text{CNOT}$

here H is acting on the target qubit

$H^\dagger H = I$

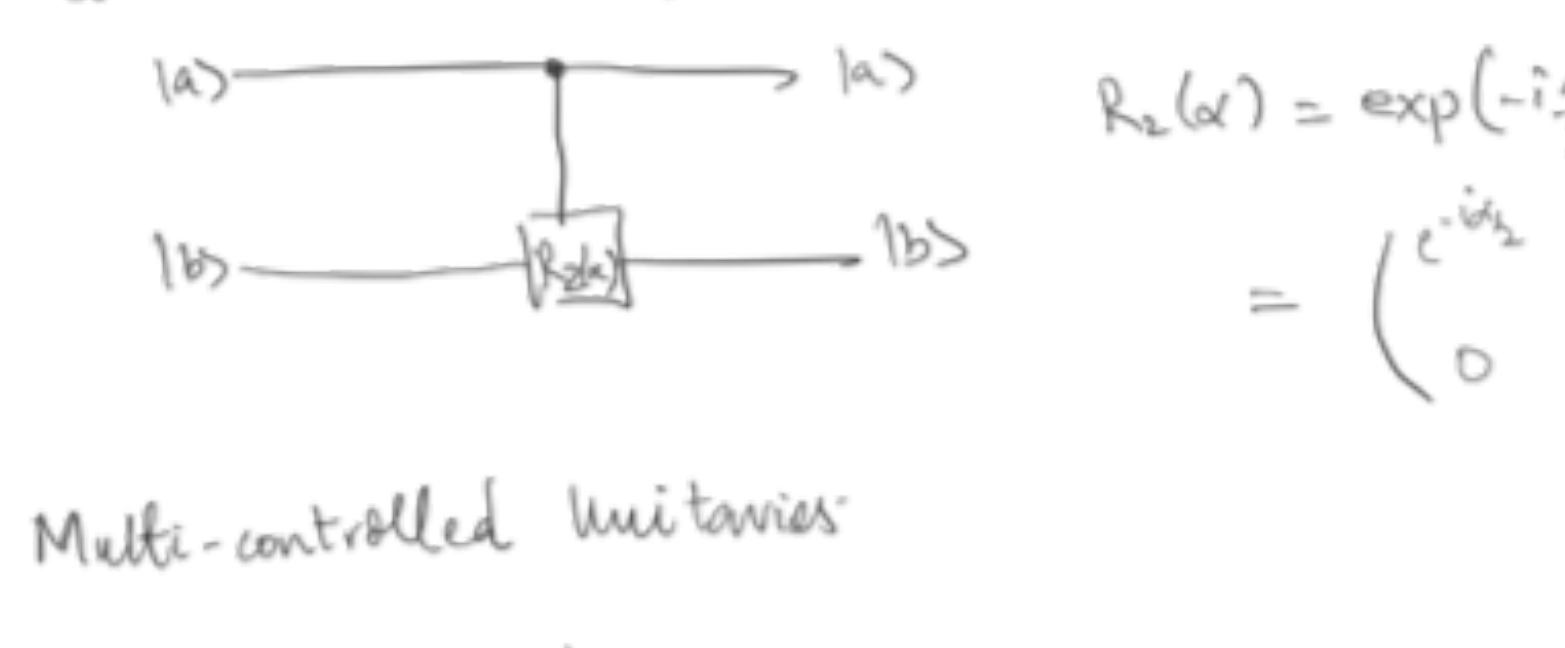
$H^2 = I$

$H^\dagger = H$

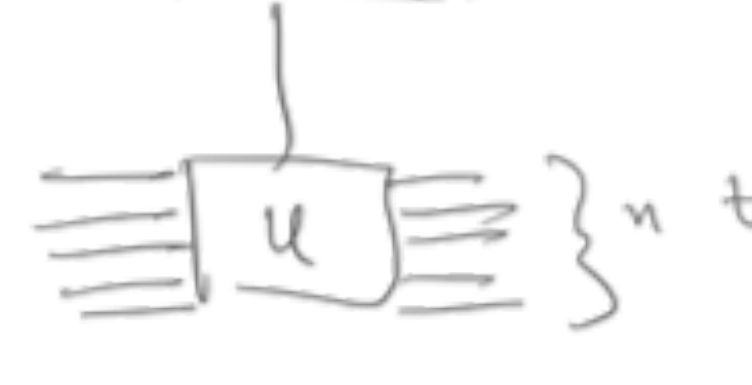
$(I \otimes H)(P_0 \otimes I + P_1 \otimes Z) I \otimes H$

$= P_0 \otimes HIH + IP_1 I \otimes HZH$

$= P_0 \otimes I + P_1 \otimes X = \text{CNOT}$



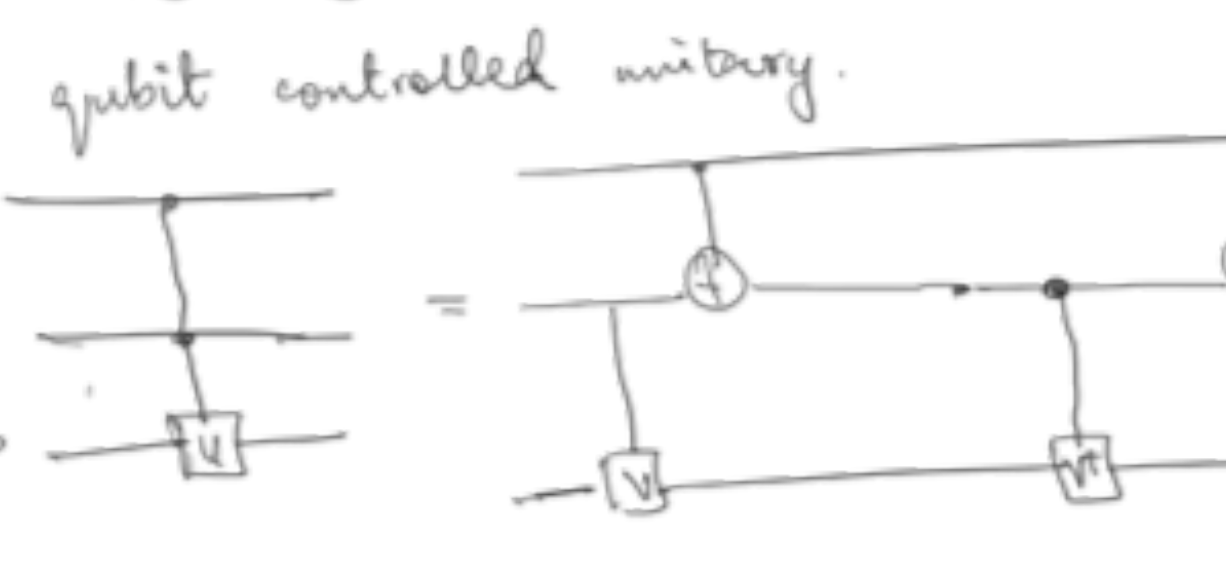
Controlled Phase Shift $R_z(\alpha)$



$R_z(\alpha) = \exp\left(-i\frac{\alpha}{2} Z\right)$

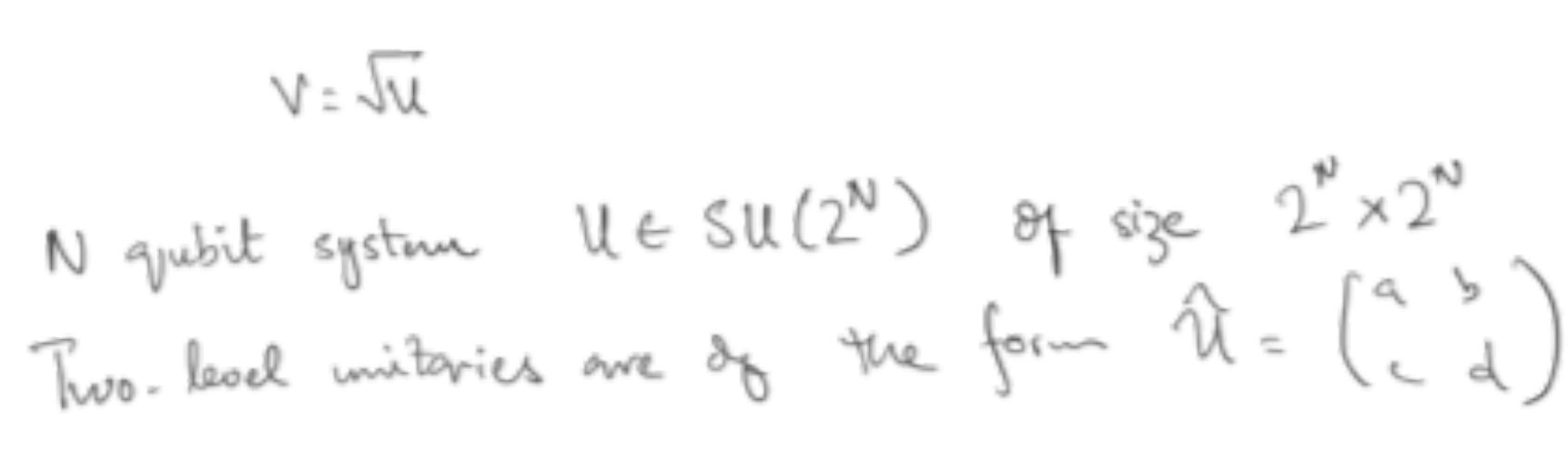
$= \begin{pmatrix} e^{-i\alpha/2} & 0 \\ 0 & e^{i\alpha/2} \end{pmatrix}$

Multi-controlled Unitaries



$P_0^1 \otimes \cdots P_0^2 \otimes \cdots P_0^m \otimes I \otimes \cdots I^{(n)} + P_1^1 \otimes P_1^2 \otimes \cdots P_1^m \otimes U^{(n)}$

Two qubit controlled unitary.



$V^2 = U$

$V = \sqrt{U}$

N qubit system $U \in SU(2^N)$ of size $2^N \times 2^N$

Two-level unitaries are of the form $\hat{U} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

U for N-qubit system

then there exists $U_1, U_2 \ldots U_d$ (two-level unitaries)

such that

$\underbrace{U_d U_{d-1} \cdots U_2 U_1} U = I$

$U^\dagger = U_d U_{d-1} U_{d-2} \cdots U_1$

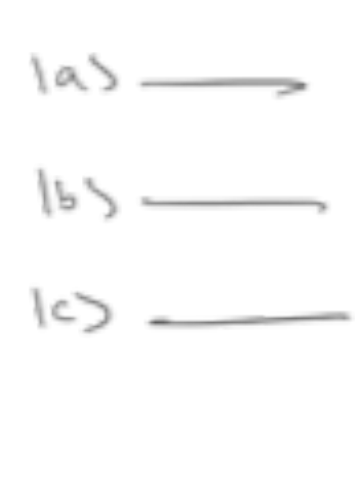
where each U_i is a two-level unitary matrix.

Two level unitaries are universal.

Any two-level unitary can be built from single qubit operations and CNOT gate.

A simple example

$U = \begin{pmatrix} a & & & & & & & c \\ & 1 & & & & & & \\ & & 1 & & & & & \\ & & & 1 & & & & \\ & & & & 1 & & & \\ & & & & & 1 & & \\ & & & & & & 1 & \\ b & & & & & & & d \end{pmatrix}$ ✓ 8×8



U acts non-trivially on $\overline{|000\rangle}$ and $\overline{|111\rangle}$

$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

$\downarrow$

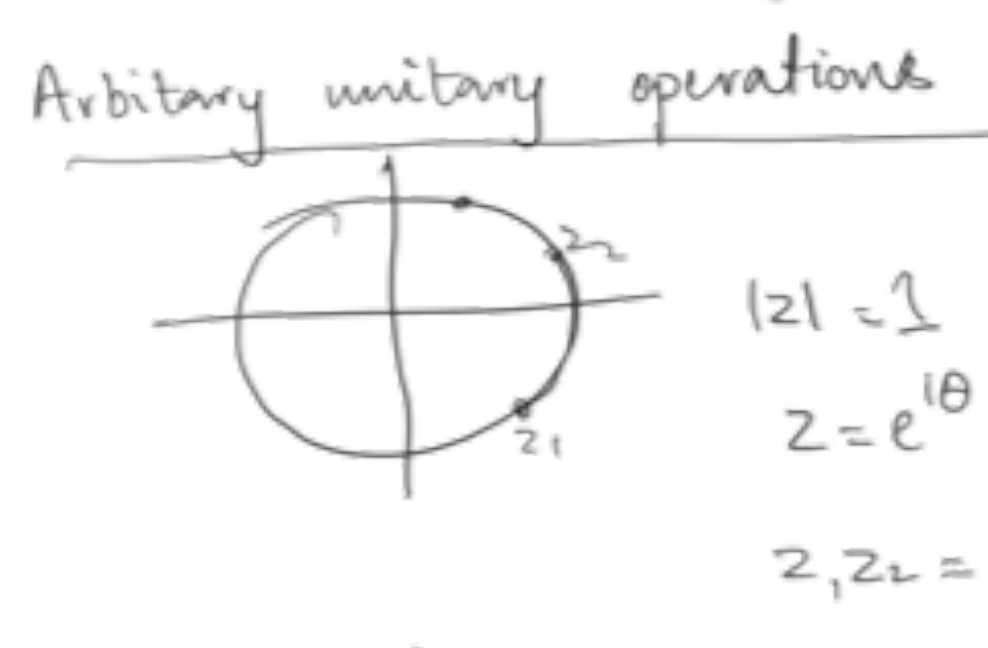
$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$

U is a two-level unitary

Write a gray code.

A	B	C
0	0	0
0	0	1
0	1	1
1	1	1

$|000\rangle$ swap it with $|011\rangle$ $\quad \hat{U} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$



$|000\rangle \rightarrow |011\rangle$ conditioned on 2nd and 3rd qubit

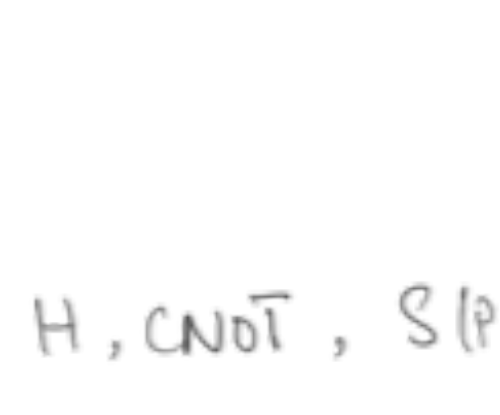
$|111\rangle$

Finally, we undo the swapping.

$(C_1 C_2)^\dagger U C_1 C_2 = \tilde{U}$

$U = (C_1 C_2)^\dagger \tilde{U} (C_1 C_2)$

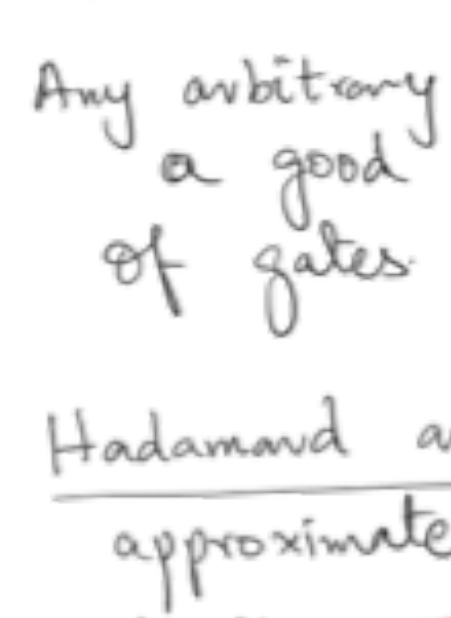
Arbitary unitary operations are hard to approximate.



$|z| = 1$

$z = e^{i\theta}$

$z_1 z_2 = z_3 = e^{i\theta_1} e^{i\theta_2} = e^{i\theta_1 + \theta_2}$



$w_1 w_N$

Discretization can't approximate continuous manifolds

$SU(2^N)$

discrete set of gates

H, CNOT, S(Phase) $\pi/8$ (T) are universal.

Any arbitrary unitary can be approximated to a good accuracy using these universal set of gates

Hadamard and $\pi/8$ (T) gate can be used to approximate any arbitrary single qubit operation.

~~Consider~~

$\boxed{\exp\left(-i\frac{\pi}{8} Z\right) \exp\left(-i\frac{\pi}{8} X\right)} = \cos^2\left(\frac{\pi}{8}\right) I - i\left(\cos\frac{\pi}{8}(X+Z) + \sin\left(\frac{\pi}{8}\right) Y\right) \sin\frac{\pi}{8}$

$\cos^2\left(\frac{\pi}{8}\right)$.

$\exp\left(i\frac{\theta}{2}\, \hat{n} . \vec{\sigma}\right) = \cos\left(\frac{\theta}{2}\right) I - i\sin\frac{\theta}{2}\left(n_x X + n_y Y + n_z Z\right)$

$\hat{n} = \left(\cos\frac{\pi}{8}, \sin\frac{\pi}{8}, \cos\left(\frac{\pi}{8}\right)\right)$

$\boxed{\cos\left(\theta/2\right) = \cos^2\left(\pi/8\right)}$

θ can be written as an irrational multiple of π.

$\underline{CXC = X \otimes X}$

$\underline{(P_0 \otimes I + P_1 \otimes X)(I \otimes X)(P_0 \otimes I + P_1 \otimes X) = X \otimes X}$

$C(X \otimes X)C = I \otimes X$

⇒ CNOT gate maps Pauli strings to Pauli strings.

Pauli group: P_n

is the n-qubit group of tensor products of Paulis with factor $\pm 1, \pm i$

$P \in P_n$

$P = i^k (P_1 \otimes P_2 \otimes P_3 \otimes \ldots P_n)$

where each $P_i = \{I, X, Y, Z\}$

4^n possibilities. $k \in 0, 1, 2, 3$

4^n strings.

4^{n+1} elements in this group.

Two qubits.

$\{II, IX, IY, IZ, ZI, YI, XI, XX, YY, ZZ, XY, YX, ZX, XZ \ldots\}$.

Clifford group: C_n

C_n is a group of unitary transformations that normalizes P_n

$\forall\, C \in C_n \quad \forall\, P \in P_n$

$\exists\, P' \in P_n$

such that $CPC^\dagger = P'$

$S^2 = Z \qquad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$

$X = HZH \qquad X = HS^2H$

$Y = SXS^\dagger \qquad S^4 = I$

$Y = SXS^3$

$Y = SHS^2HS^3$

$\{\underline{H}, \underline{S}\}$ generates $P_1 = \{\underline{I, X, Y, Z}\}$.

$= \{\underline{S^4, HS^2H, SHS^2HS^3, S^2}\}$

$T^2 = S$

$\underline{\{H, T\} \text{ also generates } P_1}$

Similarly $C_1 = \{H, T, P_1\}$.

C_n includes elements of Pauli group H, S, CNOT gate

P_n can be generated by $\{\underline{H}, S, \text{CNOT}\}$

H, S(Phase), CNOT, $\pi/8$ form a family of gates from which an arbitrary unitary can be "approximated".